

VFD Series 9000 · Overview

The Series 9000 Variable Frequency Drive is a three-phase inverter rated for 0.75 kW to 22 kW motors. It supports Modbus RTU over RS-485 and provides 12 configurable digital inputs and 4 analog outputs.

This manual covers fault codes, the Modbus register map, and commissioning steps for the standard variant (part number VFD9000-22A-S).

Fault Codes

Code	Name	Description
0x10	Undervoltage	DC bus below 380 V threshold.
0x21	Overcurrent	Output current exceeded 150% of rated for 1 s.
0x22	Overvoltage	DC bus exceeded 820 V.
0x33	Overtemperature	Heatsink above 85 C.
0x40	Ground Fault	Detected earth leakage on phase output.

Modbus Register Map (Holding Registers)

Address	Name	Type	Access
0x2000	Run Command	U16	R/W
0x2001	Output Current	U16	R
0x2002	Output Frequency	U16	R
0x2003	DC Bus Voltage	U16	R
0x2010	Acceleration Time	U16	R/W
0x2011	Deceleration Time	U16	R/W

Commissioning Steps

1. Verify mains voltage is within 380-480 V AC before applying power.
2. Connect motor leads to U, V, W terminals; observe rotation.
3. Set parameter P0.01 to motor rated current in amps.
4. Set parameter P0.02 to motor rated frequency (50 or 60 Hz).
5. Run autotune via keypad: MENU > AUTOTUNE > START.
6. Verify fault code register reads 0x00 (no fault) after autotune.

For Schneider charging station integration, refer to the EV9000 manual.